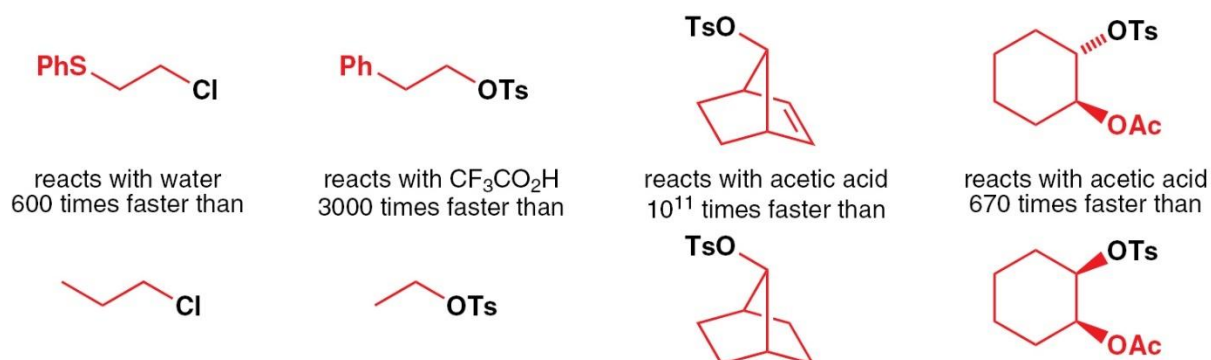
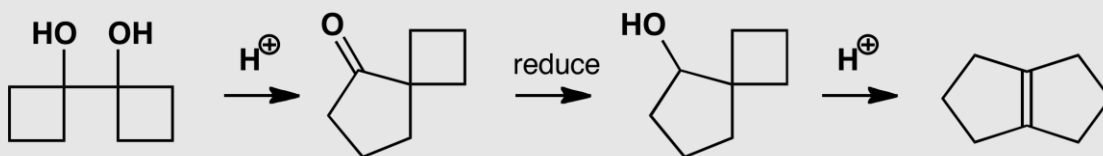


1. 写出下列底物和乙酸钠发生反应的反应机理



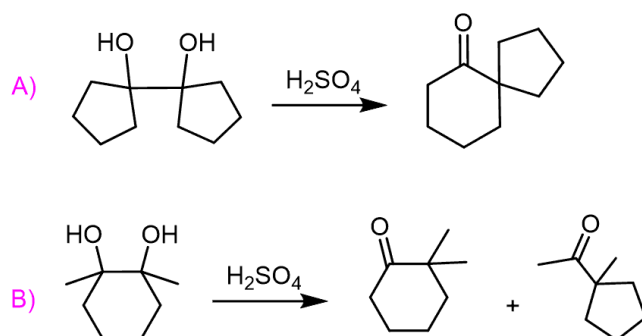
2.

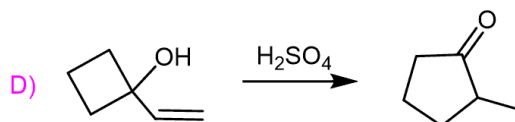
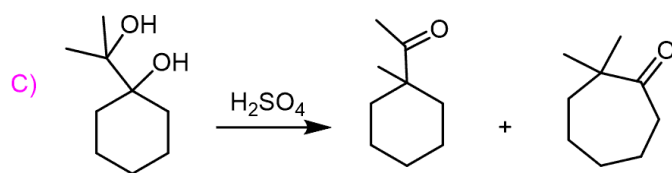
The 'pinacol' dimer of cyclobutanone rearranges with expansion of one of the rings in acid solution to give a cyclopentanone fused *spiro* to the remaining four-membered ring. Draw a mechanism for this reaction. Reduction of the ketone gives an alcohol that rearranges to a bicyclic alkene also in acid. Suggest a mechanism for this reaction and suggest why the rearrangements happen.



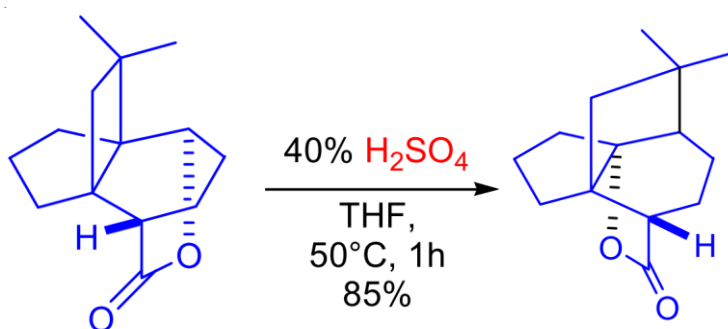
3.

Give mechanisms for these reactions that explain any selectivity.



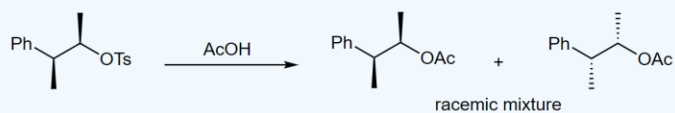


4. Draw mechanisms of the reactions below.



5

Propose a mechanism to explain the results of the following reaction. Note: You are starting with a single enantiomer of the starting material.



6.

The reactions of the isomeric starting materials produce very different products. Propose a mechanism to explain Reaction A. Why can't the starting material in Reaction B undergo a similar reaction? Reaction B is a preview of what we will see in the fragmentation chapter. Propose a mechanism to explain the product formation.

